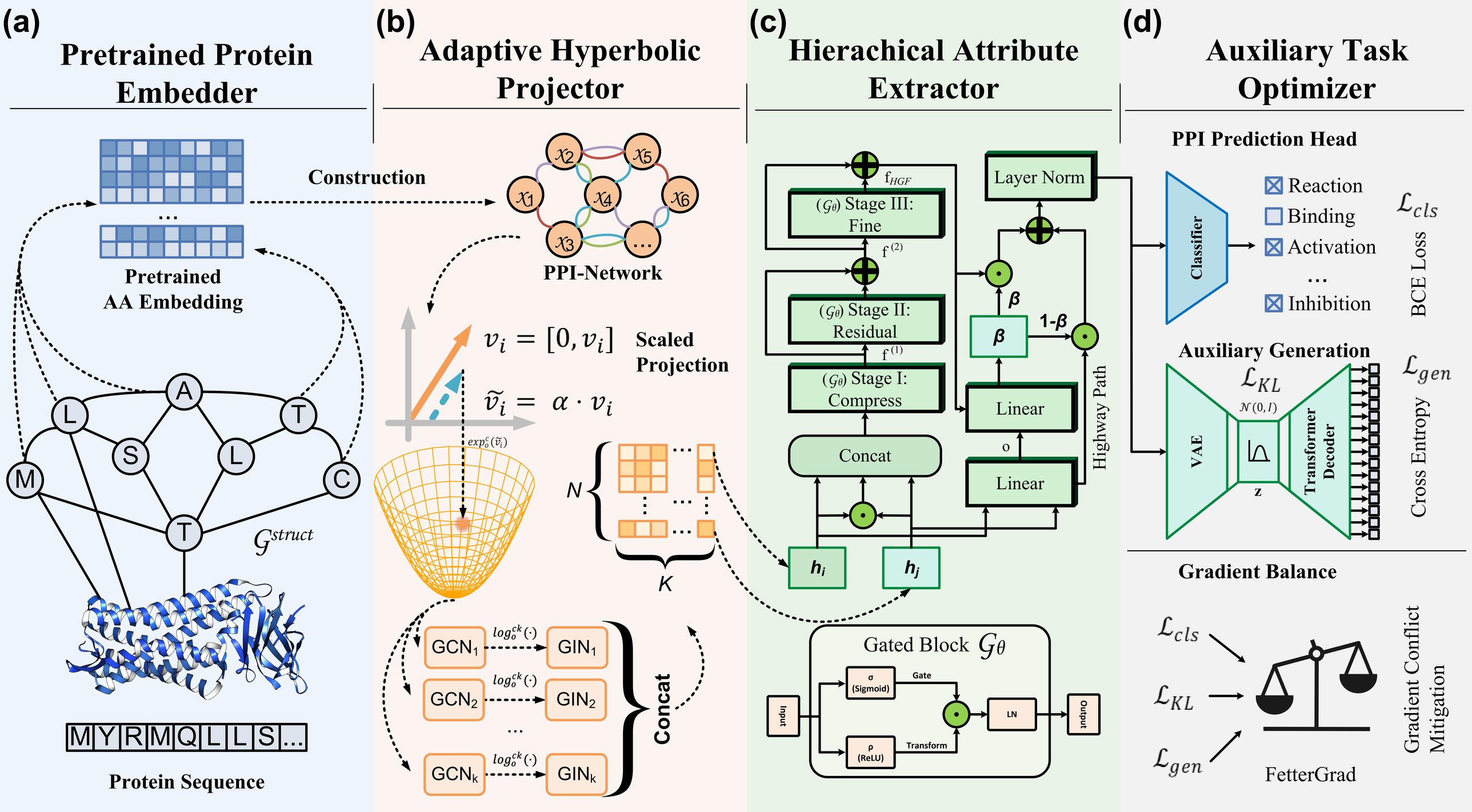
(a)
Pretrained Protein Embedder
Pretrained AA Embedding
Construction
$\mathcal{G}^{struct}$
M Y R M Q L L S ...
Protein Sequence
(b)
Adaptive Hyperbolic Projector
PPI-Network
$v_i = [0, v_i]$
Scaled Projection
$\tilde{v}_i = \alpha \cdot v_i$
$exp_o^c(\tilde{v}_i)$
N
K
GCN1
GCN2
GCNk
GIN1
GIN2
GINk
$log_o^{c_k}(\cdot)$
Concat
(c)
Hierachical Attribute Extractor
f_{HGF}
($\mathcal{G}_\theta$) Stage III: Fine
$f^{(2)}$
($\mathcal{G}_\theta$) Stage II: Residual
$f^{(1)}$
($\mathcal{G}_\theta$) Stage I: Compress
Concat
h_i
h_j
Layer Norm
β
1-β
Linear
o
Linear
Highway Path
Gated Block $\mathcal{G}_\theta$
Input
σ (Sigmoid)
Gate
ρ (ReLU)
Transform
LN
Output
(d)
Auxiliary Task Optimizer
PPI Prediction Head
Classifier
Reaction
Binding
Activation
...
Inhibition
$\mathcal{L}_{cls}$
BCE Loss
Auxiliary Generation
$\mathcal{L}_{KL}$
$\mathcal{N}(0, I)$
VAE
z
Transformer Decoder
$\mathcal{L}_{gen}$
Cross Entropy
Gradient Balance
$\mathcal{L}_{cls}$
$\mathcal{L}_{KL}$
$\mathcal{L}_{gen}$
FetterGrad
Gradient Conflict Mitigation